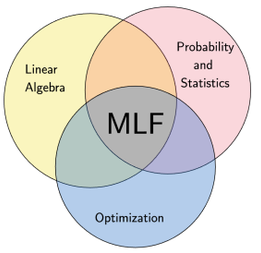
Linear Algebra Recap





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1. Machine Learning and Linear Algebra

ML $\rightarrow$ Learning from **Data**

Linear algebra gives a certain representation to data

- vectors
- matrices

$$D = \begin{bmatrix} & \text{rooms} & \text{area} & \text{distance} \\ \text{house-1} & 2 & 1000 & 5 \\ \dots & & & \\ \text{house} - 100 & 3 & 3000 & 1 \end{bmatrix}$$

We call this a **data-matrix**. Each row is a data-point. Each column is one feature for that data-point.

Data is being manipulated by ML algorithms to get some insight into it.

- Matrix multiplications
- Other manipulations

Non-linearities: activation functions (sigmoid, tanh, ReLU)

2. Column Vectors and Row Vectors

Usually vectors are denoted as $(1, 2, 0, -1) \in \mathbb{R}^4$. There are two ways of representing this:

- Column vector: $u = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix} \rightarrow 4 \times 1$
- Row vector: $u^T = \begin{bmatrix} 1 & 2 & 0 & -1 \end{bmatrix} \rightarrow 1 \times 4$

The convention is to work with column vectors.

3. Dot Products

The dot product converts two vectors to a scalar. The dot product formally can be written as a function $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$:

$$f(u, v) = u \cdot v$$

$$u = (1, 0, -1), \ v = (2, 1, 3)$$

$$u \cdot v = 2 + 0 - 3 = -1$$

If we follow this column-vector convention:

$$u \cdot v = u^T v$$

$$u = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \ v = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$u \cdot v = u^T v = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = -1$$

Remark: Dot product between two vectors $u, v \in \mathbb{R}^n$ is $u \cdot v$ which is commonly represented as $u^T v$.

Recall that $u^T v = v^T u$. This property is called symmetry.

4. Matrix Multiplication

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

There are four ways to multiply matrices having compatible dimensions ($m \times n$ and $n \times p$)

4.1. Dot-product

$$AB = \begin{bmatrix} 2 & 5 \\ 1 & 7 \end{bmatrix}$$

In general, the way you do this is:

$$A = \begin{bmatrix} - & a_1 & - \\ - & a_2 & - \end{bmatrix} B = \begin{bmatrix} | & | \\ b_1 & b_2 \\ | & | \end{bmatrix}$$

$$C = AB = \begin{bmatrix} a_1^T b_1 & a_1^T b_2 \\ a_2^T b_1 & a_2^T b_2 \end{bmatrix}$$

4.2. Matrix Column

Let us first try to understand how to multiply a matrix and a column vector:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}, \; b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$Ab = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \end{bmatrix}$$

Remark: The product of a matrix and a column vector is the linear combination of the columns of the matrix where the coefficients of the combination come from the column vector.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} Ab_1 & Ab_2 \end{bmatrix}$$

In general, if A is $m \times n$ and B is $n \times p$ and the columns of B are expressed as $b_1, \; \cdots, b_p$, then $AB = \begin{bmatrix} Ab_1 & \cdots & Ab_p \end{bmatrix}$.

4.3. Row Matrix

What happens if we multiply a row vector and a matrix?

$$a^T = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$a^T B = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= 1 \cdot \begin{bmatrix} 1 & 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix}$$

How do you extend this to AB?

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} a_1^T B \\ a_2^T B \end{bmatrix}$$

In general, if A is $m \times n$ and B is $n \times p$ and you represent $A = \begin{bmatrix} a_1^T \\ \dots \\ a_m^T \end{bmatrix}$, then AB can be expressed as:

$$AB = \begin{bmatrix} a_1^T B \\ \dots \\ a_m^T B \end{bmatrix}$$

Remark:

- Every column of AB is a linear combination of the columns of A.
- Every row of AB is a linear combination of the rows of B

Row picture

$$AB = \begin{bmatrix} a_1^T B \\ \cdots \\ a_m^T B \end{bmatrix}$$

Column picture

$$AB = \begin{bmatrix} Ab_1 & \cdots & Ab_p \end{bmatrix}$$

4.4. Outer product

The outer product is a way of multiplying two vectors which results in a matrix.

$$u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; v^T = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

$$uv^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$
\begin{aligned}
AB &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 0 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 5 \\ 1 & 7 \end{bmatrix}
\end{aligned}
$$

We will use this in week-7 when we study what is called the SVD.

In general, if matrix A $(m \times n)$ has n columns and matrix B $(n \times p)$ has n rows

$$
A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}, \; B = \begin{bmatrix} - & b_1^T & - \\ & \vdots & \\ - & b_n^T & - \end{bmatrix}
$$

Then

$$
AB = a_1 b_1^T + \cdots + a_n b_n^T
$$

Here, each $a_i b_i^T$ is a matrix of order $m \times p$ and is called an outer product.

uv^T is what is called a rank-1 matrix. In fact, every rank-1 matrix can be expressed as uv^T.

5. Row Reduction

Pivot: The leading non-zero entry in a row.
Row echelon form: A matrix A is said to be in row-echelon form if:

- All zeros rows are at the bottom.
- The pivot in any row lies to the right of the pivot in the previous row.
- Pivot should be 1.

$$\begin{bmatrix} \mathbf{1} & 0 & 3 & 4 \\ 0 & 0 & \mathbf{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} \mathbf{1} & 1 & 0 & 3 \\ 0 & \mathbf{1} & 0 & 1 \\ 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The row echelon form is not unique.

Reduced row echelon form: A matrix A is said to be in row-echelon form if:

- A is in row echelon form.
- The only non-zero entry in every pivot column is the pivot.

$$\begin{bmatrix} \mathbf{1} & 0 & 0 & -5 \\ 0 & 0 & \mathbf{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The reduced row echelon form is unique.

Row operations:

1. Swap rows
2. Adding a multiple of one row to another
3. Scaling a row by a non-zero constant

Row operations are reversible

6. Column Space

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 1 & 1 & 4 \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \; b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$Ax = x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

So $Ax = b$ can be interpreted in two ways:

Row picture

$$\begin{aligned} x_1 + 2x_2 + x_3 &= b_1 \\ 3x_2 + 2x_3 &= b_2 \\ x_1 + x_2 + 4x_3 &= b_3 \end{aligned}$$

Each row corresponds to one equation. There are as many equations as there are rows in A or entries in b.

Column picture

$$Ax = x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$Ax = b$ is the same as asking, what combination of the columns of A will give me the vector b:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = b$$

The LHS is some vector in the span of the columns of the matrix. What is span of the columns of a matrix? It is the set of all linear combinations of the columns of the matrix.

In words, this means:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = b$$

Is there any vector in the span of the columns of the matrix that is equal to b? If you can, then the system is consistent, in other words, it has one or more solutions. If b is not in the column space, then $Ax = b$ has no solution.

For example:

$$x_1\begin{bmatrix}1\\0\\1\end{bmatrix}+x_2\begin{bmatrix}2\\3\\1\end{bmatrix}+x_3\begin{bmatrix}1\\2\\4\end{bmatrix}=\begin{bmatrix}0\\-2\\-3\end{bmatrix}$$

$$x_1 = 1, x_2 = 0, x_3 = -1$$

7. Null space

The null space of a matrix A is the set of all solutions to $Ax = 0$. In other words, it is the set of all solutions to the homogeneous system $Ax = 0$.

$$N(A) = \{x : Ax = 0\}$$

- matrix: may-tricks
- matrices: may-tri-seez
- infinite: in-finit

8. System of Linear Equations

$$A \to m \times n$$

$$x \to n \times 1$$

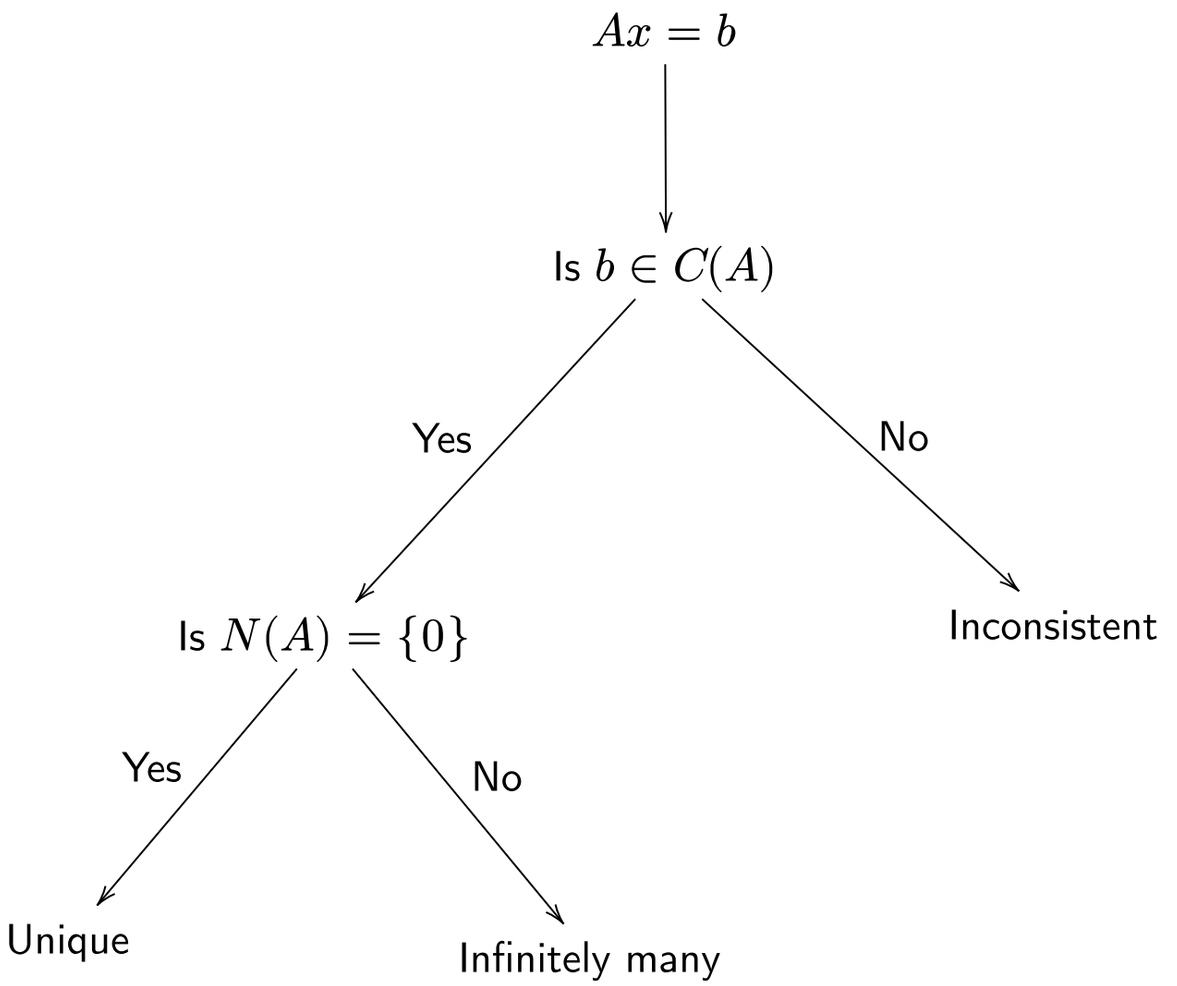
$$b \to m \times 1$$

$$Ax = b$$

What is x? Solve for x.

- $x = A^{-1}b$ is a class-12 answer.

- The answer is another question: is b in the column space of A
  - if $b \notin C(A)$, then $Ax = b$ has no solutions
  - if $b \in C(A)$, then $Ax = b$ has solutions
- If $b \in C(A)$, how many solutions does the system have
  - If $N(A) = \{0\}$, then $Ax = b$ has only one solution
  - If $N(A) \neq \{0\}$, then $Ax = b$ has infinitely many solutions



If $Ax = b$ is inconsistent, then is it at least approximately equal to b?

$$Ax \approx b$$

Least squares problem $\rightarrow$ Gauss first introduced it in the domain of astronomy

The true undergraduate response to "Solve for $Ax = b$"

$$x = A^{\dagger}b$$

Linear regression $\rightarrow$ MLT, DL

9. Transpose, Inverse, Rank and Determinant

If A is a $m \times n$ matrix, then A^T is a $n \times m$ matrix whose rows and columns have been swapped.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \; A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

Properties:

- $(AB)^T = B^T A^T$
- $\left(A^T\right)^T = A$

What is the rank of a matrix? Number of linearly independent columns/rows.

- Row rank, column rank
- Row rank and column rank are equal

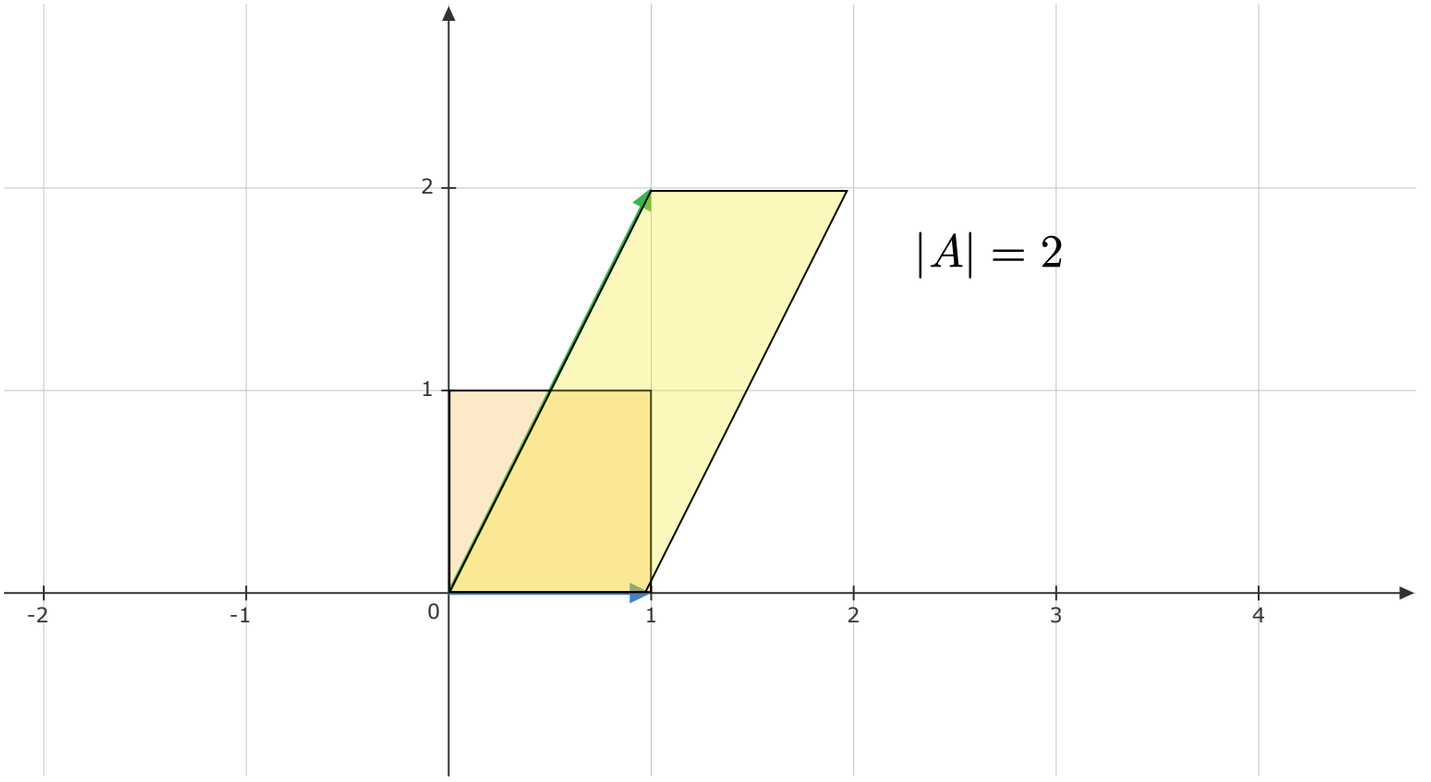
Properties of rank

- $\text{rank}(A) = \text{rank}\left(A^T\right)$
- $\text{rank}(A) \leqslant \min\left(\text{number of rows, number of columns}\right)$

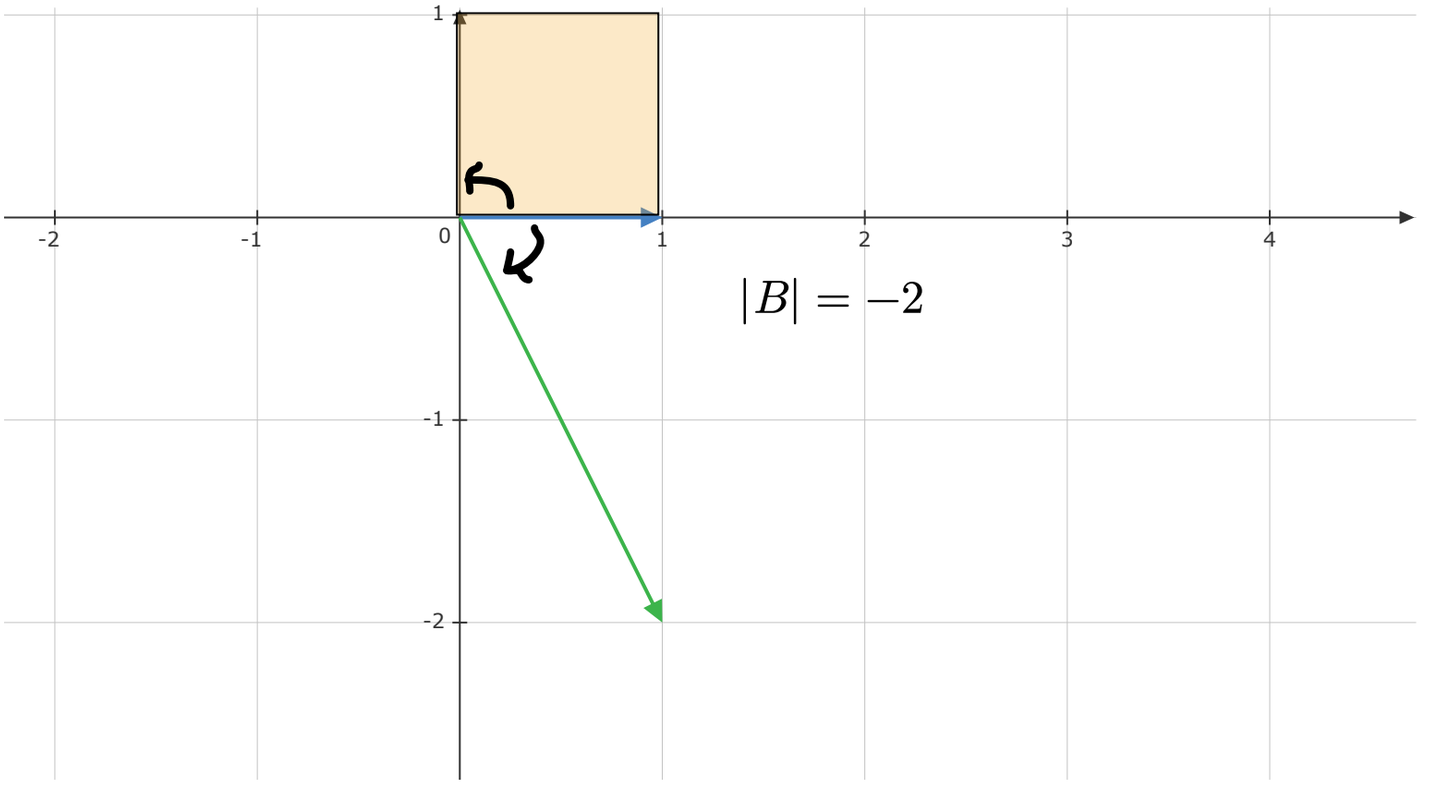
The determinant of a square matrix is a function that maps the matrix to a real number.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

What happens to the area of the square formed by the basis vectors $(1, 0)$ and $(0, 1)$ under the transformation by the matrix A? Its area is scaled by $|A|$.



$$B = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$$



Properties of determinants

- $|AB| = |A| \cdot |B| \rightarrow$ determinant is multiplicative in nature
- If two rows are equal, then the determinant is zero.
- If two rows are swapped, then the sign of the determinant changes.

A square matrix A of order n is called invertible if we can find a matrix B of the same order such that $AB = BA = I$. Then we say that $B = A^{-1}$. How many inverses does a matrix have? Unique.

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A)$$

- A is invertible if and only if $|A| \neq 0$.
- $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$
- The rank of an invertible matrix of order n is $n \rightarrow$ it is full rank

A matrix which is not full-rank is called rank-deficient matrix.

10. Some Special Square Matrices

- Diagonal: All non-diagonal entries are zero.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \; B = 0$$

- Upper Triangular: All entries below the diagonal are zero

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 10 \\ 0 & 0 & 5 \end{bmatrix}$$

Row reduction leads to one matrix decomposition called the LU decomposition

- Lower
- Upper

$$A = LU$$

- Symmetric: $A = A^T$

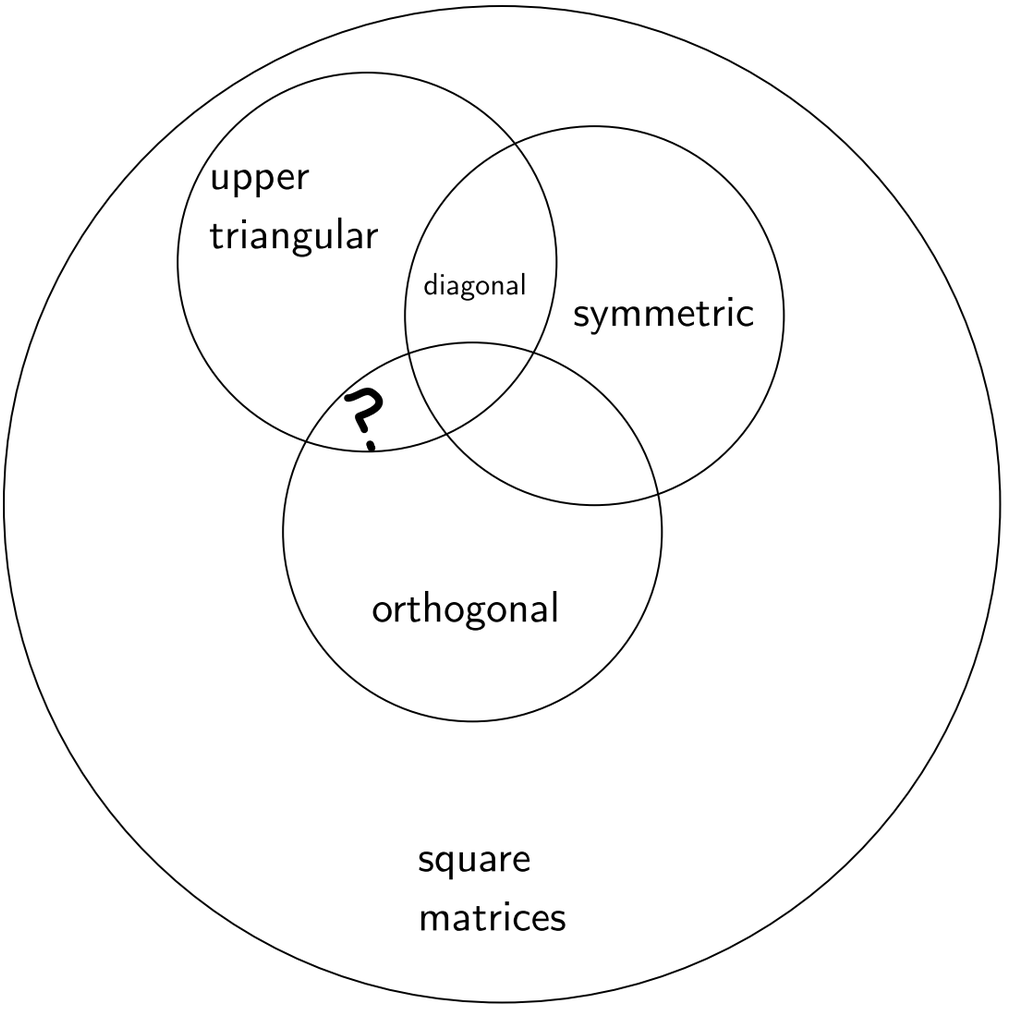
$$A_{ij} = A_{ji}$$

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 1 & 10 \\ 5 & 10 & 0 \end{bmatrix}$$

- Orthogonal
  - An orthogonal matrix has orthonormal columns.

$$Q_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},\ Q_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, Q_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

They are often seen in rotations and reflections.


upper
triangular
diagonal
symmetric
?
orthogonal
square
matrices